

Chapter 26: Bloch Sphere and Single-Qubit Arbitrary Unitary Gate

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Bloch Sphere

- Bloch Sphere is a very useful tool to understand the operation of quantum gates.
- We are all familiar with the concept that a single qubit resides in the $\mathbb{C}^2$ space. This is a two-dimensional complex space. It means that it has two basis vectors, $|0\rangle$ and $|1\rangle$, which are orthonormal to each other.
- Any qubit state is a vector in this space formed by a linear combination of $|0\rangle$ and $|1\rangle$ with complex coefficients.
- We have been using a real 2D space to illustrate the properties of a qubit state vector, but we emphasize that this is just an illustration and the $\mathbb{C}^2$ space is NOT the real 2D space that we can feel.
- How many degrees of freedom (DOF) does a qubit state have? We know that any single-qubit state $|\psi\rangle$ can be expressed as $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$, where α and β are complex numbers. Each of the α and β is determined by two real numbers. For example, $\alpha = |\alpha|e^{i\delta_\alpha}$ and $\beta = |\beta|e^{i\delta_\beta}$, where $|\alpha|$, δ_α , $|\beta|$, and δ_β are the real numbers. Therefore, it looks like has 4 DOFs.
- For any physical qubit, it has to be normalized. Therefore, $|\alpha|^2 + |\beta|^2 = 1$. This reduces the DOF by 1 because if we specify $|\alpha|$, $|\beta|$ is specified at the same time due to normalization equation. So, we can determine $|\alpha|$ and $|\beta|$ using 1 single real number parameter as long as we make the vector normalized.

Bloch Sphere

- For example, we can set $|\alpha| = \cos \frac{\theta}{2}$ and $|\beta| = \sin \frac{\theta}{2}$ as $\cos^2 \frac{\theta}{2} + \sin^2 \frac{\theta}{2} = 1$ will help satisfy the normalization criteria. So, now, the amplitudes of α and β can be described by a single real parameter θ , and the DOF of the qubit state is 3.
- The global phase means the phase shared by both $|0\rangle$ and $|1\rangle$. It has no physical significance because the physical properties for this single qubit are all computed by involving the bra and ket versions of the vector. For example, the expectation value of a matrix, $\mathbf{M}$, is $\langle\psi|\mathbf{M}|\psi\rangle$. We can take $e^{i\gamma}$ out from $|\psi\rangle$ to be $e^{i\gamma}$ and take $e^{i\gamma}$ out from $\langle\psi|$ to be $e^{-i\gamma}$. $e^{i\gamma}$ and $e^{-i\gamma}$ multiply together to be 1. Therefore, the global phase factor does not affect the physical results.
- As a result, we only need two real parameters, θ and ϕ , to describe a qubit as following:

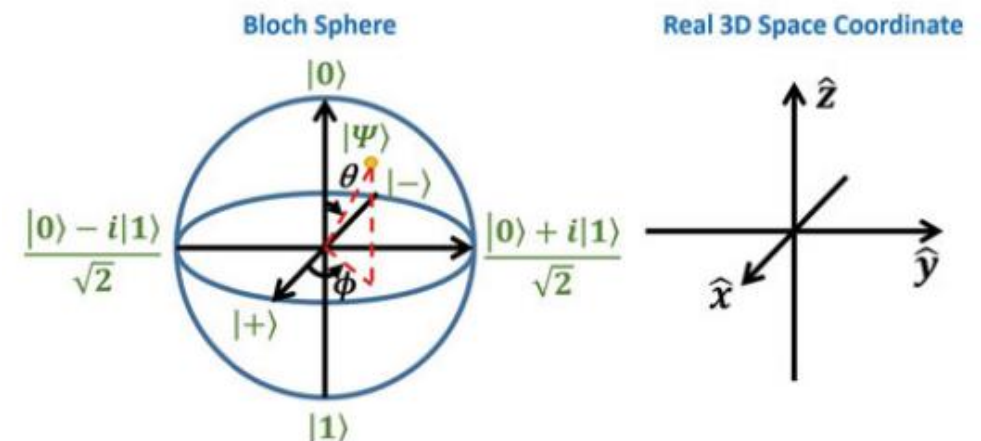
$$|\psi\rangle = \cos \frac{\theta}{2} e^{-i\phi} |0\rangle + \sin \frac{\theta}{2} e^{i\phi} |1\rangle$$

Since it has only two parameters, we can describe it on the *surface* of a unit sphere (sphere with unit length), and this is called the Bloch Sphere.

Bloch Sphere

- Pic. 41 shows the Bloch sphere and its relationship to the real 3D space we live in. The surface of the Bloch sphere can be mapped to the space of a qubit. We may say that we map the qubit space in our real 3D space.
- The Bloch sphere allows us to “visualize” how a qubit state evolves. Second, some of the physical properties of the qubit are linked to the Bloch sphere orientation in the real 3D space.
- Pic. 41 shows the location of state $|\psi\rangle$ on the Bloch sphere when it has the parameters θ and ϕ . We see that if we put the Bloch sphere in the 3D orientation shown in Pic. 41, the θ and ϕ are just the **polar angle** and **azimuthal angle** of a spherical coordinate, respectively.
- Consider the “North Pole” It has $\theta = 0$ and $\phi = 0$. Therefore,

$$\begin{aligned}
 |\psi_{\theta=0,\phi=0}\rangle &= \cos\frac{\theta}{2}e^{-i\phi/2}|0\rangle + \sin\frac{\theta}{2}e^{i\phi/2}|1\rangle \\
 &= \cos\frac{0}{2}e^{-i0/2}|0\rangle + \sin\frac{0}{2}e^{i0/2}|1\rangle \\
 &= |0\rangle
 \end{aligned}$$



Pic. 41 The Bloch sphere (left) and its relationship to the 3-D real space (right)

Bloch Sphere

- What if we choose ϕ to be non-zero? It is still the “North Pole” as long as $\theta = 0$. Then we will get $e^{-i\phi/2}|0\rangle$. However, since $e^{-i\phi/2}$ is a global phase (as the coefficient for $|1\rangle$ is 0), this does not matter to the physics.
- Remember that we ignore the global phase of the qubit state when we construct the Bloch sphere; therefore, the Bloch sphere has lost the global phase information.
- For the same point on the Bloch sphere, we can add any global phase to it without affecting its physical properties. And a point on the Bloch sphere is not unique, but it corresponds to qubits that differ in only a global phase (e.g. $e^{-i\phi/2}|0\rangle$ and $|0\rangle$ in this case).
- Consider the point corresponds to $x = -1$ and $y = z = 0$ in the 3D real space. It has $\theta = \frac{\pi}{2}$ and $\phi = \pi$. So,

$$\begin{aligned}
 |\psi_{\theta=\pi/2, \phi=\pi}\rangle &= \cos\frac{\theta}{2}e^{-i\phi/2}|0\rangle + \sin\frac{\theta}{2}e^{i\phi/2}|1\rangle \\
 &= \cos\frac{\pi}{4}e^{-i\pi/2}|0\rangle + \sin\frac{\pi}{4}e^{i\pi/2}|1\rangle \\
 &= \frac{1}{\sqrt{2}}(-i)|0\rangle + \frac{1}{\sqrt{2}}i|1\rangle \\
 &= -i\frac{1}{\sqrt{2}}(|0\rangle - |1\rangle) = -i|-\rangle
 \end{aligned}$$

Again, the global phase $-i = e^{-i\pi/2}$ can be discarded. So, this point corresponds to $|-\rangle$.

Bloch Sphere

- Consider the point corresponds to $y = 1$ and $x = z = 0$ in the 3D real space. It has $\theta = \frac{\pi}{2}$ and $\phi = \pi/2$. So,

$$\begin{aligned}
 |\psi_{\theta=\pi/2, \phi=\pi/2}\rangle &= \cos\frac{\theta}{2} e^{-i\phi/2} |0\rangle + \sin\frac{\theta}{2} e^{i\phi/2} |1\rangle \\
 &= \cos\frac{\pi}{4} e^{-i\pi/4} |0\rangle + \sin\frac{\pi}{4} e^{i\pi/4} |1\rangle \\
 &= \frac{1}{\sqrt{2}} e^{-\frac{i\pi}{4}} (|0\rangle + e^{\frac{i\pi}{2}} |1\rangle) \\
 &= e^{-i\pi/4} \frac{1}{\sqrt{2}} (|0\rangle + i|1\rangle)
 \end{aligned}$$

Again, the global phase $e^{-i\pi/4}$ can be discarded. Therefore, this point corresponds to $\frac{1}{\sqrt{2}} (|0\rangle + i|1\rangle)$, which is one of the eigenvalues σ_y .

- From the example, the extrema of the Bloch sphere in the x, y, and z directions correspond to the eigenvalues of the σ_x , σ_y , and σ_z matrices, respectively. This is the first place where we see it is related to our real 3D space. This is because, for example, if it is a spin qubit under an external magnetic field the Pauli matrices are related to the directions of the magnetic field. So now we see the link between the extrema in a direction (e.g. z direction) on the Bloch sphere to the external magnetic direction (e.g. z direction).

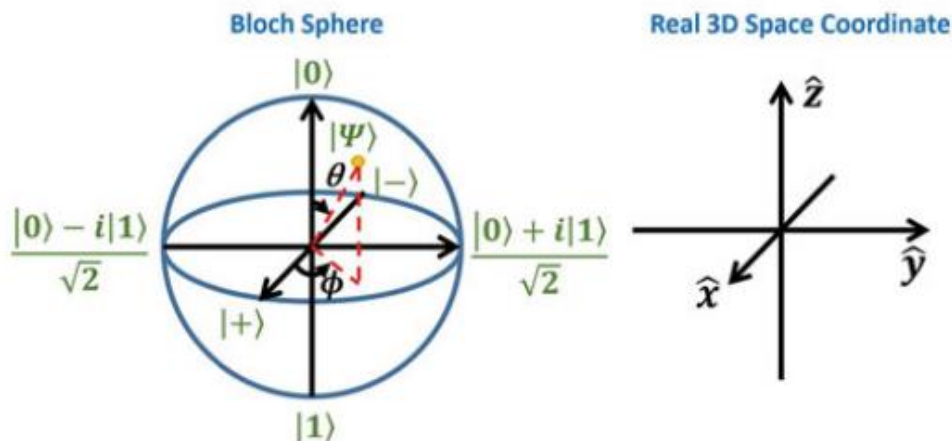
Expectation Values of Pauli Matrices

- The Pauli matrices are related to the directions of the external magnetic field in the spin qubit case.
- Moreover, the expectation values of the Pauli matrices of any state are related to the energy it gains under the external magnetic field. Therefore, the calculation of the expectation value of the Pauli matrices is very important.
- Find the expectation value of σ_z of $|\psi\rangle$. First, we can express $|\psi\rangle$ in its column form. That is $|\psi\rangle = \cos \frac{\theta}{2} e^{-i\phi/2} |0\rangle + \sin \frac{\theta}{2} e^{i\phi/2} |1\rangle = \begin{pmatrix} \cos \frac{\theta}{2} e^{-i\phi/2} \\ \sin \frac{\theta}{2} e^{i\phi/2} \end{pmatrix}$. Therefore,

$$\begin{aligned}
 \langle \psi | \sigma_z | \psi \rangle &= \left(\cos \frac{\theta}{2} e^{\frac{i\phi}{2}} \quad \sin \frac{\theta}{2} e^{-\frac{i\phi}{2}} \right) \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} \cos \frac{\theta}{2} e^{-i\phi/2} \\ \sin \frac{\theta}{2} e^{i\phi/2} \end{pmatrix} \\
 &= \left(\cos \frac{\theta}{2} e^{\frac{i\phi}{2}} \quad \sin \frac{\theta}{2} e^{-\frac{i\phi}{2}} \right) \begin{pmatrix} \cos \frac{\theta}{2} e^{-i\phi/2} \\ -\sin \frac{\theta}{2} e^{i\phi/2} \end{pmatrix} \\
 &= \cos \frac{\theta}{2} e^{\frac{i\phi}{2}} \cos \frac{\theta}{2} e^{-\frac{i\phi}{2}} + \sin \frac{\theta}{2} e^{-\frac{i\phi}{2}} \left(-\sin \frac{\theta}{2} e^{\frac{i\phi}{2}} \right) \\
 &= \cos \frac{\theta}{2} \cos \frac{\theta}{2} - \sin \frac{\theta}{2} \sin \frac{\theta}{2} = \cos \theta
 \end{aligned}$$

Expectation Values of Pauli Matrices

- What is the geometric meaning of $\cos \theta$ on the Bloch sphere in Pic. 41. Since it is a unit sphere, this is just the projection of the state $|\psi\rangle$ on the z-axis!
- Although the states on the Bloch sphere have no direct relationship to the real 3D space, their Paul matrices' expectation values turn out to be the projections on the axes from the states on the Bloch sphere.
- Therefore, although $|0\rangle$ and $|1\rangle$ do not lie in our real 3D space on the opposite sides of the z-axis, their σ_z expectation values do have opposite values. These expectation values are related to the energy splitting and magnetic moments of the qubits.
- We can also calculate the expectation values of σ_x and σ_y , and they are just the projections of the state on the x- and y-axis, respectively. If so, without any derivation, based on the geometry, we know that



$$\langle \psi | \sigma_x | \psi \rangle = \sin \theta \cos \phi$$

$$\langle \psi | \sigma_y | \psi \rangle = \sin \theta \sin \phi$$

Pic. 41 The Bloch sphere (left) and its relationship to the 3-D real space (right)

Single-Qubit Arbitrary Unitary Rotation

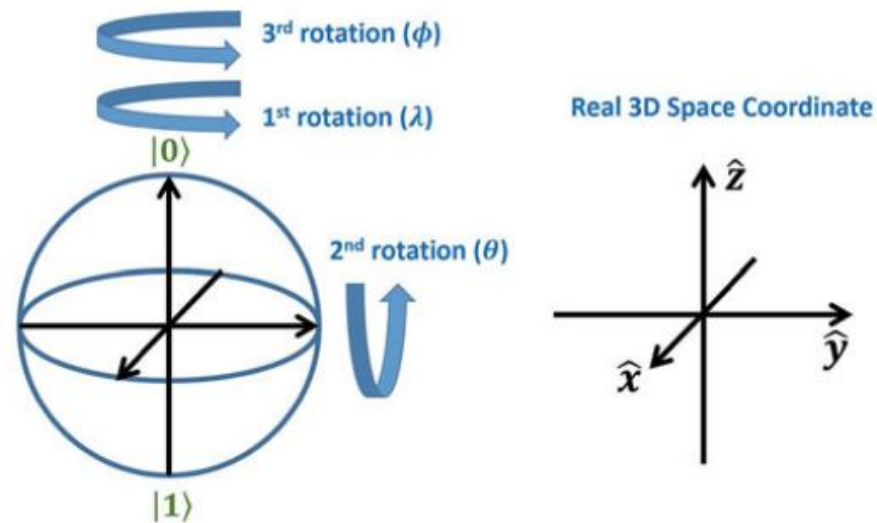
- We have been emphasizing that quantum computing is nothing but just the rotation of the quantum states in the hyperdimensional space that we cannot see. For example, applying a NOT gate to the $|0\rangle$ state will rotate it to the $|1\rangle$ state.
- Note that the rotation on the Bloch sphere is NOT the rotation of the state in the 3D space. The qubit space is not something we can feel. This is just a convenient visualization.
- How do we describe this rotation? This rotation is a rotation in the hyperspace where the qubit state resides, and it must be a unitary rotation and it must be a 2×2 matrix. In general, it is described by a **Single-Qubit Arbitrary Unitary Gate** $U_{\theta, \phi, \lambda}$.

$$U_{\theta, \phi, \lambda} = \begin{pmatrix} \cos \frac{\theta}{2} & -e^{i\lambda} \sin \frac{\theta}{2} \\ e^{i\phi} \sin \frac{\theta}{2} & e^{i(\lambda+\phi)} \cos \frac{\theta}{2} \end{pmatrix}$$

- The parameters, θ , ϕ , and λ , are angles, and they are the **Euler angles** in the **Euler rotation**.

Single-Qubit Arbitrary Unitary Rotation

- Euler rotation is a sequence of three rotations. It can be described in two ways. One is the *extrinsic rotation* that it rotates about fixed 3D coordinate axes (Pic. 42).
- In the extrinsic rotation, the body will first rotate about the z -axis by λ (also called γ in some literature) and then rotate about the y -axis by θ (also called β in some literature) and finally rotate about the z -axis by ϕ (also called α in some literature).
- Also be careful that in the $U_{\theta, \phi, \lambda}$ notation, we do not put the rotation angles in the order of rotation (i.e. θ, ϕ, λ instead of ϕ, θ, λ or λ, θ, ϕ).



Pic. 42 Relationship between the Bloch sphere and Euler rotation

Single-Qubit Arbitrary Unitary Rotation

- Construct a NOT gate using $U_{\theta, \phi, \lambda}$ by matching the matrices as follows:

$$U_{\theta, \phi, \lambda} = U_{NOT} \\ \begin{pmatrix} \cos \frac{\theta}{2} & -e^{i\lambda} \sin \frac{\theta}{2} \\ e^{i\phi} \sin \frac{\theta}{2} & e^{i(\lambda+\phi)} \cos \frac{\theta}{2} \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

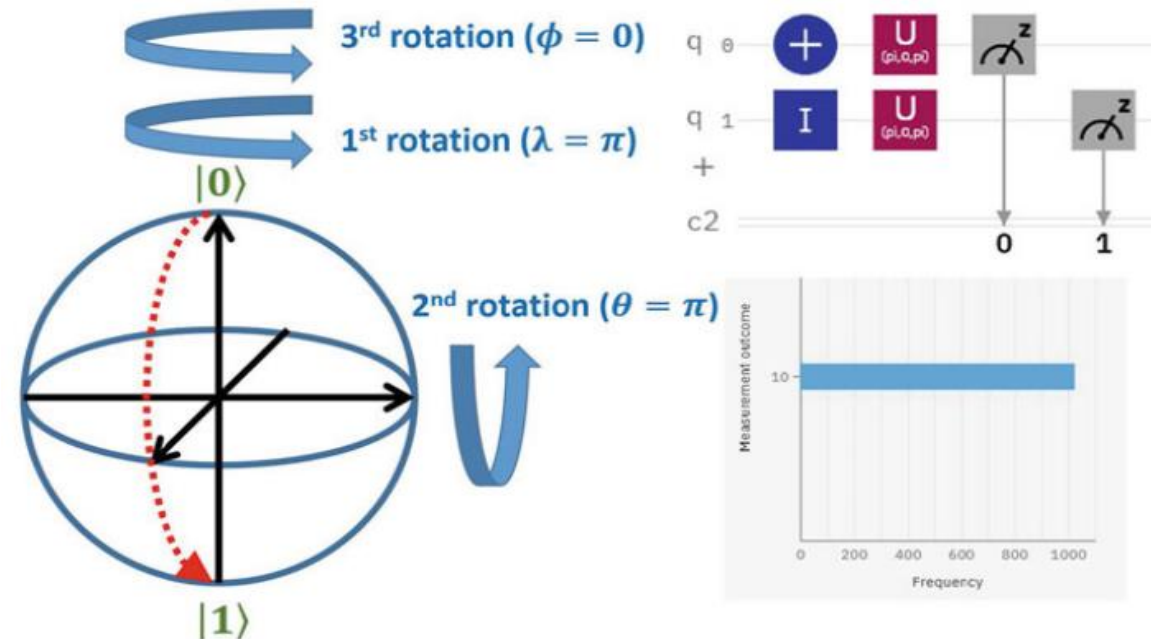
- We can set up four equations to match each of the elements.

$$\begin{aligned} \cos \frac{\theta}{2} &= 0 \\ -e^{i\lambda} \sin \frac{\theta}{2} &= 1 \\ e^{i\phi} \sin \frac{\theta}{2} &= 1 \\ e^{i(\lambda+\phi)} \cos \frac{\theta}{2} &= 0 \end{aligned}$$

- $\theta = \pi$ is required to make the diagonal element zero as $\cos \frac{\pi}{2} = 0$. Then $\sin \frac{\theta}{2} = \sin \frac{\pi}{2} = 1$. And we can put $\lambda = \pi$ so that $-e^{i\lambda} = 1$ and $\phi = 0$ so that $e^{i\phi} = 1$. So, the solution is $(\theta, \phi, \lambda) = (\pi, 0, \pi)$.

Single-Qubit Arbitrary Unitary Rotation

- Pic. 43 shows the path of the rotation of the $|0\rangle$ state under this quantum gate.
- When it is at the “North” or “South” poles, the first and third rotations have no effects.



Pic. 43 Implementation of U_{NOT} using $U_{\theta, \phi, \lambda}$. Left shows how $|0\rangle$ is rotated on the Bloch sphere under a NOT operation. Right shows the implementation on IBM-Q with $|01\rangle$ as the input. MSB is at the bottom of the circuit

- We can see the second rotation (about y-axis by $\theta = \pi$) brings it to $|1\rangle$. The MSB (bottom) has an input of $|0\rangle$, and the LSB (top) has an input of $|1\rangle$ (after a NOT gate) to the $U_{\theta, \phi, \lambda} = U_{\pi, 0, \pi}$. Therefore, the input is $|01\rangle$, and we can see the output is $|10\rangle$ as expected and, thus, $U_{\pi, 0, \pi}$ behave as a U_{NOT} .

Single-Qubit Arbitrary Unitary Rotation

- Construct the matrix corresponding to the rotation of $|0\rangle$ to $|1\rangle$ on the Bloch sphere.
- Pic. 43 clearly shows that the initial state $|0\rangle$ is at the “North” pole and the final state is at the “South” pole.
- Rotating about the y -axis by π in the second rotation and do nothing in the first and third rotations. This corresponds to $U_{\theta, \phi, \lambda} = U_{\pi, 0, 0}$. Then we obtain

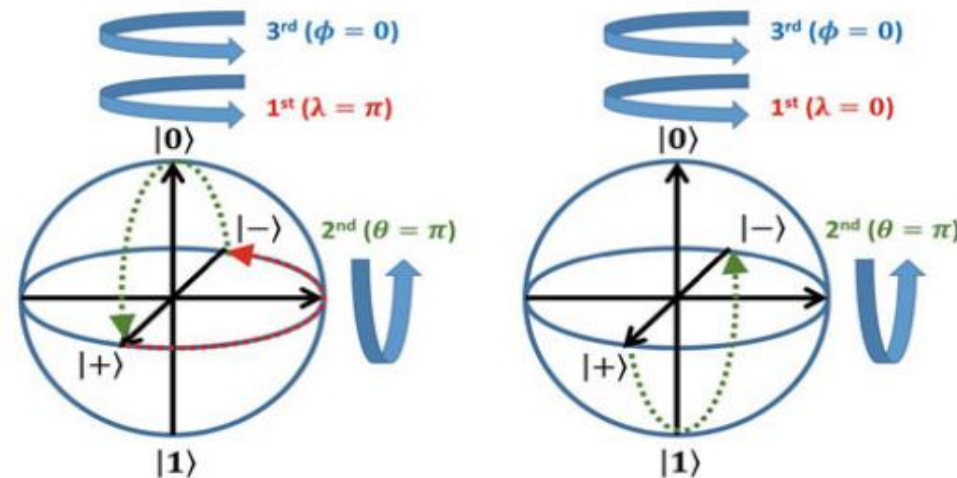
$$\begin{aligned}
 U_{\pi, 0, 0} &= \begin{pmatrix} \cos \frac{\theta}{2} & -e^{i\lambda} \sin \frac{\theta}{2} \\ e^{i\phi} \sin \frac{\theta}{2} & e^{i(\lambda+\phi)} \cos \frac{\theta}{2} \end{pmatrix} \\
 &= \begin{pmatrix} \cos \frac{\pi}{2} & -e^{i\lambda} \sin \frac{\pi}{2} \\ e^{i\phi} \sin \frac{\pi}{2} & e^{i(0+0)} \cos \frac{\pi}{2} \end{pmatrix} \\
 &= \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}
 \end{aligned}$$

Unfortunately, this is not a NOT gate. But this matrix correctly describes the transformation from $|0\rangle$ to $|1\rangle$ because

$$U_{\pi, 0, 0}|0\rangle = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \end{pmatrix} = |1\rangle$$

Single-Qubit Arbitrary Unitary Rotation

- For example, describe how $|+\rangle$ rotates on the Bloch sphere when the NOT gate is applied.
- First, we know that $\mathbf{U}_{NOT}|+\rangle = \mathbf{U}_{NOT} \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) = \frac{1}{\sqrt{2}}(|1\rangle + |0\rangle) = |+\rangle$. We used the definition of $\mathbf{U}_{NOT}$ that it changes $|0\rangle$ and $|1\rangle$ to each other. So, $|+\rangle$ is transformed/rotated to itself.
- The NOT gate is $\mathbf{U}_{\pi,0,\pi}$ on the Bloch sphere, it will first rotate about the z -axis by π , and then about the y -axis by π , and do nothing in the third rotation as $\phi = 0$.
- Pic. 44 shows that it is brought to $|-\rangle$ in the first rotation and brought back to $|+\rangle$ in the second rotation. In the same figure, it shows that if we use $\mathbf{U}_{\pi,0,0}$, we get the wrong result because $\mathbf{U}_{\pi,0,0}$ is not the NOT gate.



Pic. 44 Left: Rotation path of $|+\rangle$ under a NOT gate operation. Right: Rotation path of $|+\rangle$ under the $\mathbf{U}_{\pi,0,0}$ operation

Chapter 27:

Quantum Phase Estimation

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General Controlled Unitary Gate

- The single-qubit arbitrary unitary gate, $U_{\theta,\phi,\lambda}$ describes how a single qubit rotates on the Bloch sphere.
- We discarded the global phase information because for a single qubit, the global phase has no physical significance.
- The rotation on a Bloch sphere can be described by the $U_{\theta,\phi,\lambda}$ defined as following:

$$U_{\theta,\phi,\lambda} = \begin{pmatrix} \cos \frac{\theta}{2} & -e^{i\lambda} \sin \frac{\theta}{2} \\ e^{i\phi} \sin \frac{\theta}{2} & e^{i(\lambda+\phi)} \cos \frac{\theta}{2} \end{pmatrix}$$

This gate represents all possible one-qubit gates as long as we do not care about the global phase of the state. But it is a problem when multiple qubits are considered.

- When more qubits are involved, while the global phase of the whole system does not matter, the “global phases” of the individual qubits matter.
- The global phases of the individual qubits determine the relative phase between different qubits and can result in constructive or destructive interferences that are important mechanisms in quantum computing.

General Controlled Unitary Gate

- Therefore, *when we are not allowed to neglect the global phase of a single qubit*. We can extend the concept, so it will be able to represent any unitary rotation of a qubit by adding a phase factor, γ , as the following:

$$U_{\theta, \phi, \lambda, \gamma} = e^{i\gamma} \begin{pmatrix} \cos \frac{\theta}{2} & -e^{i\lambda} \sin \frac{\theta}{2} \\ e^{i\phi} \sin \frac{\theta}{2} & e^{i(\lambda+\phi)} \cos \frac{\theta}{2} \end{pmatrix}$$

The θ , ϕ , and λ still have the same meaning as in $U_{\theta, \phi, \lambda}$.

- We can see that when we only have $U_{\theta, \phi, \lambda}$, for some matrices, we are missing the $e^{i\gamma}$. This causes an error when it is applied to a vector by omitting $e^{i\gamma}$. But for a single-qubit case, missing a phase does not matter.

General Controlled Unitary Gate

- For example, represent the gate $\begin{pmatrix} i & 0 \\ 0 & -i \end{pmatrix}$ in the form of $\mathbf{U}_{\theta, \phi, \lambda, \gamma}$.

$$\begin{pmatrix} i & 0 \\ 0 & -i \end{pmatrix} = i \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} = e^{i\pi/2} \begin{pmatrix} \cos \frac{0}{2} & -e^{i\pi} \sin \frac{0}{2} \\ e^{i0} \sin \frac{0}{2} & e^{i(\pi+0)} \cos \frac{0}{2} \end{pmatrix}$$

- Therefore, $\mathbf{U}_{0, 0, \pi, \pi/2}$ is the correct representation. When we apply this gate to $\begin{pmatrix} \alpha \\ \beta \end{pmatrix}$, we get

$$\mathbf{U}_{0, 0, \pi, \pi/2} \begin{pmatrix} \alpha \\ \beta \end{pmatrix} = \begin{pmatrix} i & 0 \\ 0 & -i \end{pmatrix} \begin{pmatrix} \alpha \\ \beta \end{pmatrix} = i \begin{pmatrix} \alpha \\ -\beta \end{pmatrix}$$

- If we use $\mathbf{U}_{0, 0, \pi}$ instead, a phase factor of $e^{i\gamma} = e^{i\pi/2} = i$ is missing. When we apply $\mathbf{U}_{0, 0, \pi}$ to $\begin{pmatrix} \alpha \\ \beta \end{pmatrix}$, it becomes

$$\mathbf{U}_{0, 0, \pi} \begin{pmatrix} \alpha \\ \beta \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} \alpha \\ \beta \end{pmatrix} = \begin{pmatrix} \alpha \\ -\beta \end{pmatrix}$$

The rotated state vectors are the same except differ by the phase factor, i .

General Controlled Unitary Gate

- To construct a general version of a controlled unitary gate. Since two qubits are involved, we cannot ignore the global phase of each qubit anymore.
- Therefore, the controlled unitary gate must use all four parameters, and its symbol is $\mathcal{C} - U_{\theta, \phi, \lambda, \gamma}$.
- It has this matrix:

$$\mathcal{C} - U_{\theta, \phi, \lambda, \gamma} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & e^{i\gamma} \cos \frac{\theta}{2} & -e^{i(\lambda+\gamma)} \sin \frac{\theta}{2} \\ 0 & 0 & e^{i(\phi+\gamma)} \sin \frac{\theta}{2} & e^{i(\lambda+\phi+\gamma)} \cos \frac{\theta}{2} \end{pmatrix}$$

- Here we assume the MSB is the control qubit and the LSB is the target qubit. We can find this matrix by using the definition. A $\mathcal{C} - U$ gate means that if the control qubit in the basis vector is 0, it does nothing to the target qubit, and if the control qubit is 1, it will apply the unitary gate. Therefore, the definition can be written as:

$$\begin{aligned} \mathcal{C} - U_{\theta, \phi, \lambda, \gamma} |00\rangle &= I|0\rangle \otimes I|0\rangle = |00\rangle \\ \mathcal{C} - U_{\theta, \phi, \lambda, \gamma} |01\rangle &= I|0\rangle \otimes I|1\rangle = |01\rangle \\ \mathcal{C} - U_{\theta, \phi, \lambda, \gamma} |10\rangle &= I|1\rangle \otimes U_{\theta, \phi, \lambda, \gamma} |0\rangle \\ \mathcal{C} - U_{\theta, \phi, \lambda, \gamma} |11\rangle &= I|1\rangle \otimes U_{\theta, \phi, \lambda, \gamma} |1\rangle \end{aligned}$$

General Controlled Unitary Gate

$$\begin{aligned}
 \mathcal{C} - U_{\theta, \phi, \lambda, \gamma} |00\rangle &= I|0\rangle \otimes I|0\rangle = |00\rangle \\
 \mathcal{C} - U_{\theta, \phi, \lambda, \gamma} |01\rangle &= I|0\rangle \otimes I|1\rangle = |01\rangle \\
 \mathcal{C} - U_{\theta, \phi, \lambda, \gamma} |10\rangle &= I|1\rangle \otimes U_{\theta, \phi, \lambda, \gamma} |0\rangle \\
 &= |1\rangle \left(e^{i\gamma} \cos \frac{\theta}{2} |0\rangle + e^{i(\phi+\gamma)} \sin \frac{\theta}{2} |1\rangle \right) \\
 &= e^{i\gamma} \cos \frac{\theta}{2} |10\rangle + e^{i(\phi+\gamma)} \sin \frac{\theta}{2} |11\rangle \\
 \mathcal{C} - U_{\theta, \phi, \lambda, \gamma} |11\rangle &= I|1\rangle \otimes U_{\theta, \phi, \lambda, \gamma} |1\rangle \\
 &= |1\rangle \left(-e^{i(\lambda+\gamma)} \sin \frac{\theta}{2} |0\rangle + e^{i(\lambda+\phi+\gamma)} \cos \frac{\theta}{2} |1\rangle \right) \\
 &= -e^{i(\lambda+\gamma)} \sin \frac{\theta}{2} |10\rangle + e^{i(\lambda+\phi+\gamma)} \cos \frac{\theta}{2} |11\rangle
 \end{aligned}$$

- E.g. find the element ($e_{2,2}$) of the second row and second column (counting from 0-th) of $\mathcal{C} - U_{\theta, \phi, \lambda, \gamma}$. The element corresponds to $\langle 10|$ (row) and $|10\rangle$ (column). Using the above equation:

$$e_{2,2} = \langle 10 | \mathcal{C} - U_{\theta, \phi, \lambda, \gamma} | 10 \rangle = e^{i\gamma} \cos \frac{\theta}{2}$$

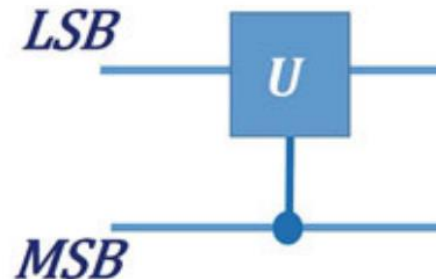
- When we apply $\langle 10|$ and we use the orthonormal property of the basis vectors that $\langle 10 | 10 \rangle = 1$ and $\langle 10 | 11 \rangle = 0$. This is the same element $e_{2,2}$ as in the matrix $\mathcal{C} - U_{\theta, \phi, \lambda, \gamma}$

General Controlled Unitary Gate

- Finally, we may also construct the matrix $\mathbf{C} = U_{\theta, \phi, \lambda, \gamma}$ using this equation:

$$\mathbf{C} = U_{\theta, \phi, \lambda, \gamma} = |0\rangle\langle 0| \otimes I + |1\rangle\langle 1| \otimes U_{\theta, \phi, \lambda, \gamma}$$

- It is easy to see the result if we recognize that $|0\rangle\langle 0| = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$ and $|1\rangle\langle 1| = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}$. We then perform the tensor products accordingly.
- This equation says that we should perform identity operation to the target qubit if the control qubit is 0 and apply the unitary gate to the target qubit if the control qubit is 1.
- Pic. 45 shows the gate symbol.



Pic. 45 Symbol of a $\mathbf{C} = U_{\theta, \phi, \lambda, \gamma}$ gate with the MSB as the control qubit

Quantum Phase Estimation

- The purpose of the **Quantum Phase Estimation (QPE)** algorithm is to estimate the phase, $2\pi\psi$, of the eigenvalue of a unitary $m \times m$ matrix, U .
- Since it is a unitary matrix, its eigenvalue must have a magnitude of one. This is because we can always transform the matrix into a basis formed by its eigenvectors (as the basis states), and its diagonal elements are just its eigenvalues.
- And since the columns of the unitary matrix must be normalized, then the eigenvalues must have a magnitude of one and be in the form of $e^{i2\pi\psi}$. We choose to write its phase as $2\pi\psi$ instead of a single variable is just for convenience in the derivation.
- Let us look at a QPE example for a 2×2 matrix. Assume we are given a phase shift gate with $\phi = \pi$, $U_{PS,\pi}$ which is just a Z-gate, Z .

$$U_{PS,\pi} = \begin{pmatrix} 1 & 0 \\ 0 & e^{i\pi} \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} = \begin{pmatrix} e^{i2\pi 0} & 0 \\ 0 & e^{i2\pi \frac{1}{2}} \end{pmatrix}$$

We see that it is already in its diagonal form, and therefore, the elements are its eigenvalues, 1 and -1 , which have phases of 0 and π , respectively. We express them in the form of $e^{i2\pi\psi}$, $\psi_0 = 0$ and $\psi_1 = 1/2$.

Quantum Phase Estimation

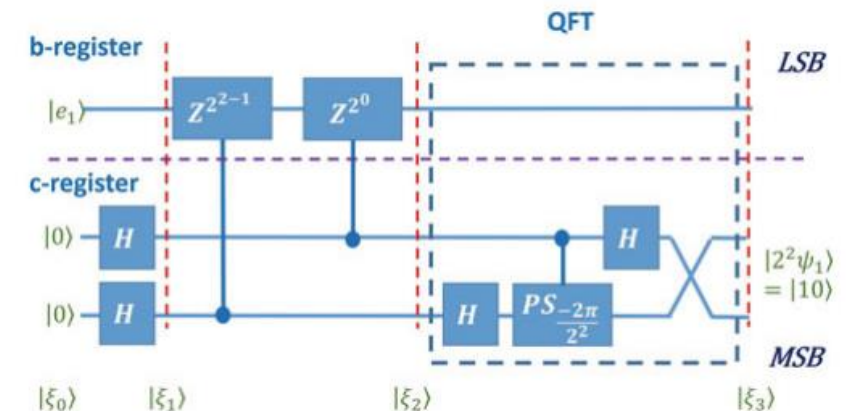
- Each of the eigenvalues corresponds to an eigenvector, and they are $|e_0\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$ and $|e_1\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$, respectively.
- We can prove that they are the correct eigenvectors by showing, by definition,

$$U_{PS,\pi}|e_0\rangle = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = 1 \begin{pmatrix} 1 \\ 0 \end{pmatrix} = 1|e_0\rangle$$

$$U_{PS,\pi}|e_1\rangle = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} 0 \\ 1 \end{pmatrix} = -1 \begin{pmatrix} 0 \\ 1 \end{pmatrix} = -1|e_1\rangle$$

The goal of the QPE is to find ψ_0 and ψ_1 for the given $U_{PS,\pi}$.

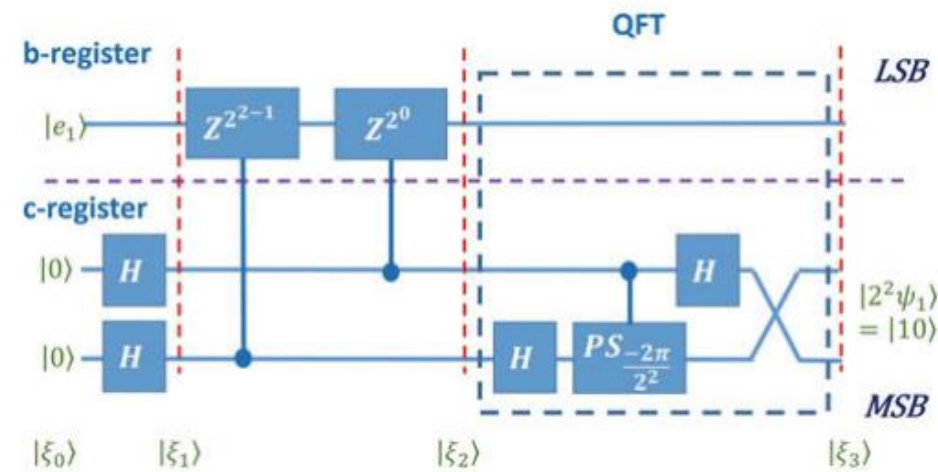
- Pic 46. shows the quantum circuit for finding the phase of the Z-gate. To be exact, it finds ψ instead of $2\pi\psi$.
- There are two groups of inputs. The less significant qubits (called the b-register) take the eigenvector of the Z-gate. Since the Z-gate is a 2×2 matrix, we only need 1-qubit to represent its eigenvector.
- In the example, we assume the input is $|e_1\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$. The other group of inputs (called the c-register) contains the more significant qubits. In this example, we use two qubits.



Pic. 46 The QPE circuit for finding the phase of the second eigenvalue of the Z-gate ($U_{PS,\pi}$). Note that the bottom is the MSB

Quantum Phase Estimation

- These qubits will be used to store a value related to ψ , which is $2^n\psi$, where n is the number of the qubits in the c-register. In this case, it is $2^n\psi = 4\psi$.
- The inputs to the c-register are the ground state $|0\rangle_2$. This is a $2+1 = 3$ -qubit circuit. Therefore, it starts with $|\xi_0\rangle = |0\rangle|0\rangle|e_1\rangle$.



Pic. 46 The QPE circuit for finding the phase of the second eigenvalue of the Z-gate (UPS, π). Note that the bottom is the MSB

- The circuit begins with a two-dimensional Hadamard gate ($H \otimes H$) to create a superposition on the c-register.

Quantum Phase Estimation

$$\begin{aligned}
 |\xi_0\rangle &= H \otimes H \otimes I (|0\rangle|0\rangle|e_1\rangle) \\
 &= H|0\rangle \otimes H|0\rangle \otimes I|e_1\rangle \\
 &= \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)|e_1\rangle \\
 &= \frac{1}{2}(|00\rangle + |01\rangle + |10\rangle + |11\rangle)|e_1\rangle \\
 &= \frac{1}{2}(|00\rangle|e_1\rangle + |01\rangle|e_1\rangle + |10\rangle|e_1\rangle + |11\rangle|e_1\rangle)
 \end{aligned}$$

- After that, controlled gates are used. The type of controlled gate used is related to the Z-gate, of which we are finding the eigenvalue phase. If we were to find that of another gate/unitary matrix (e.g. the NOT gate), we need to use the controlled gate related to that new gate (e.g. CNOT gate).
- It is related to the Z-gate because, besides controlled-Z-gate, it also uses controlled- Z^k gates, where $k = 2^l$, with l running from 0 to $n - 1$, n is the number of qubits in the c-register.
- A controlled- Z^k gate means that, for a basis state, if the control qubit is 0, it does nothing to the target qubit. If the control qubit is 1, it will apply Z^k gate to the target qubit.
- Each qubit in the c-register acts as the control bit of one of the controlled gates. A Z^k gate is equivalent to applying Z-gate k times. In this case, l runs from 0 to 1 and thus k can be 1 or 2.

Quantum Phase Estimation

- What happens to the b-register qubit when the controlled- Z^k gates are applied? Note that the b-register has the eigenvector of $\mathbf{Z}$.
- Therefore, the operation will give the eigenvalue of $\mathbf{Z}$ because of $\mathbf{U}_{PS,\pi}|e_1\rangle = \mathbf{Z}|e_1\rangle = e^{i2\pi\frac{1}{2}}|e_1\rangle = -1|e_1\rangle$.
- And if it is $\mathbf{Z}^2$, it becomes $(e^{i2\pi\frac{1}{2}})^2|e_1\rangle = |e_1\rangle$ as $\mathbf{Z}$ is applied twice. However, this only happens if the corresponding control qubit is 1. Therefore, the n controlled gates produce the following results:

$$\begin{aligned}
 |\xi_2\rangle &= \frac{1}{2}(|00\rangle|e_1\rangle + |01\rangle\mathbf{Z}|e_1\rangle + |10\rangle\mathbf{Z}^2|e_1\rangle + |11\rangle\mathbf{Z}^2\mathbf{Z}|e_1\rangle) \\
 &= \frac{1}{2}(|00\rangle\mathbf{Z}^0\mathbf{Z}^0|e_1\rangle + |01\rangle\mathbf{Z}^0\mathbf{Z}^1|e_1\rangle + |10\rangle\mathbf{Z}^2\mathbf{Z}^0|e_1\rangle + |11\rangle\mathbf{Z}^2\mathbf{Z}^1|e_1\rangle) \\
 &= \frac{1}{2}(|00\rangle\mathbf{Z}^{0+0}|e_1\rangle + |01\rangle\mathbf{Z}^{0+1}|e_1\rangle + |10\rangle\mathbf{Z}^{2+0}|e_1\rangle + |11\rangle\mathbf{Z}^{2+1}|e_1\rangle) \\
 &= \frac{1}{2}(|0\rangle\mathbf{Z}^0|e_1\rangle + |1\rangle\mathbf{Z}^1|e_1\rangle + |2\rangle\mathbf{Z}^2|e_1\rangle + |3\rangle\mathbf{Z}^3|e_1\rangle) \\
 &= \frac{1}{2} \sum_{j=0}^3 |j\rangle\mathbf{Z}^j|e_1\rangle
 \end{aligned}$$

- In the first line, $\mathbf{Z}$ is applied to the b-register if qubit 0 of the c-register is one, and $\mathbf{Z}^2$ is applied to the b-register if qubit 1 of the c-register is one. This is based on the definition of the controlled gates.
- In the second line, we also add $\mathbf{Z}^0$ that is just an identity gate when the control qubits are 0. In the third line, we group the $\mathbf{Z}$'s so we can see the sum of their exponents for each basis vector. And in the fourth line, we write the basis vectors in decimal.
- So, the action of all controlled gates is equivalent to applying $\mathbf{Z}^j$ gate on the b-register if the c-register is $|j\rangle$.¹⁹⁸

Quantum Phase Estimation

- And since $|e_1\rangle$ is an eigenvector of $\mathbf{Z}^j$, $\mathbf{Z}^j |e_1\rangle = (e^{i2\pi\frac{1}{2}})^j |e_1\rangle$. So,

$$\begin{aligned} |\xi_2\rangle &= \frac{1}{2} \sum_{j=0}^3 |j\rangle \mathbf{Z}^j |e_1\rangle \\ &= \frac{1}{2} \sum_{j=0}^3 |j\rangle (e^{i2\pi\frac{1}{2}})^j |e_1\rangle \\ &= \frac{1}{2} \sum_{j=0}^3 (e^{i2\pi\frac{1}{2}})^j |j\rangle |e_1\rangle \end{aligned}$$

- The next step is to apply QFT. What really important is that we need to use the matrix for rotating the vector instead of the basis and the matrix is shown in following:

$$U_{QFT} = \frac{1}{\sqrt{N}} \begin{pmatrix} \omega^{-0.0} & \dots & \omega^{-0.(N-1)} \\ \vdots & \ddots & \vdots \\ \omega^{-(N-1).0} & \dots & \omega^{-(N-1).(N-1)} \end{pmatrix}$$

- The definition of QFT is repeated here,

$$U_{QFT} |j\rangle = \frac{1}{\sqrt{N}} \sum_{k=0}^{N-1} \omega^{-kj} |k\rangle$$

Quantum Phase Estimation

- We only apply it to the c-register and keep the b-register intact. Therefore, we will apply $U_{QFT} \otimes I$ to $|\xi_2\rangle$, and we have the right-hand side equation:
- The first three lines just apply U_{QFT} to the c-register in $|\xi_2\rangle$, use the linear property of quantum mechanics, and then apply U_{QFT} .
- The last two lines use the definition of the n -th root of unity, ω .
- Finally, we group the terms. Note that the b-register is no longer useful now. *It is completely disentangled with the rest of the qubit, and it can be factorized out in the tensor product.*
- If we perform a measurement on the c-register, we will get one of the $|k\rangle$'s. However, if we rewrite the equation by shuffling the summations, we have

$$|\xi_3\rangle = \frac{1}{4} \sum_{j=0}^3 \left(\sum_{k=0}^3 (\omega)^{j(2-k)} \right) |k\rangle |e_1\rangle$$

- We see that each $|k\rangle$ has a coefficient of $\sum_{j=0}^3 (\omega)^{j(2-k)}$, and this is a summation of the n -th root of unity. This term is 0 unless $(2 - k) = 0$ due to the **constructive interference**.

$$\begin{aligned} |\xi_3\rangle &= U_{QFT} \otimes I \left(\frac{1}{2} \sum_{j=0}^3 (e^{i2\pi\frac{1}{2}})^j |j\rangle |e_1\rangle \right) \\ &= \frac{1}{2} \sum_{j=0}^3 (e^{i2\pi\frac{1}{2}})^j (U_{QFT} |j\rangle) \otimes I |e_1\rangle \\ &= \frac{1}{2} \sum_{j=0}^3 (e^{i2\pi\frac{1}{2}})^j \left(\frac{1}{\sqrt{4}} \sum_{k=0}^3 (\omega)^{-kj} |k\rangle \right) |e_1\rangle \\ &= \frac{1}{4} \sum_{j=0}^3 \sum_{k=0}^3 (e^{i2\pi\frac{1}{2}})^j \omega^{-kj} |k\rangle |e_1\rangle \\ &= \frac{1}{4} \sum_{j=0}^3 \sum_{k=0}^3 (e^{i2\pi\frac{2}{4}})^j (e^{i2\pi/4})^{-kj} |k\rangle |e_1\rangle \\ &= \frac{1}{4} \sum_{j=0}^3 \sum_{k=0}^3 (\omega)^{j(2-k)} |k\rangle |e_1\rangle \end{aligned}$$

Quantum Phase Estimation

- Therefore, if we perform the measurement, we will only measure $|k\rangle$, if $(2 - k) = 0$. The basis state we will measure is always

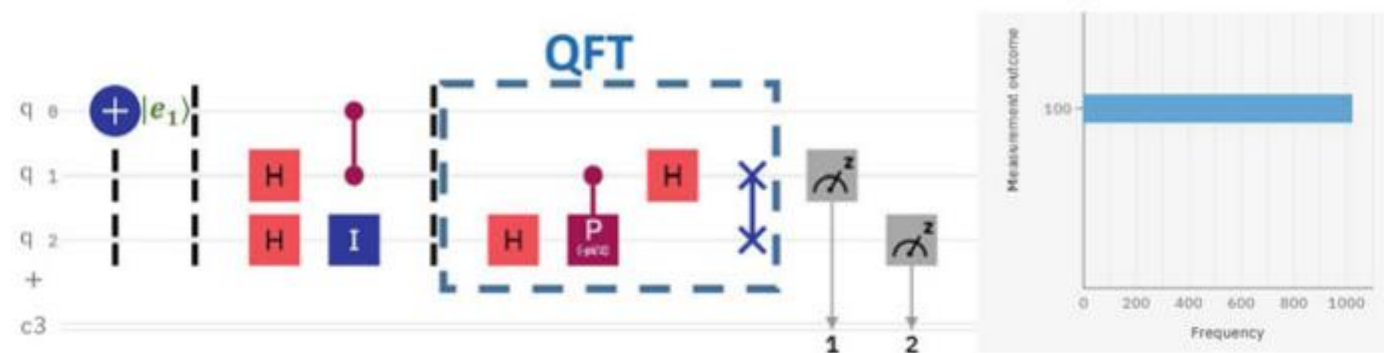
$$|k\rangle = |2\rangle_{10} = |10\rangle_2$$

- What is the meaning of $k = 2$? Let us inspect carefully how we obtain $\omega^{j(2-k)}$ in previous equation. It came from $(e^{i2\pi\frac{1}{2}})^j \omega^{-kj}$. The “2” in $\omega^{j(2-k)}$ is obtained by multiplying $\frac{1}{2}$ by 4. And 4 is the $N = 2^2$ in the N -th root of unity. Note that $\frac{1}{2}$ is the ψ in the eigenvalue of eigenvector $|e_1\rangle$, and 4 is $2^n = 2^2$, where n is the number of qubits in the c-register. Therefore, the $k = 2$ we obtained is due to the $\psi(\frac{1}{2})$ being multiplied by $2^n (2^2 = 4)$.
- If we measure $|k\rangle$ in a n -qubit c-register, and if the input to the b-register is the eigenvector corresponding to the eigenvalue, $e^{i2\pi\psi}$, the phase is given by

$$2\pi\psi = \frac{2\pi k}{2^n}$$

Implementation on IBM-Q

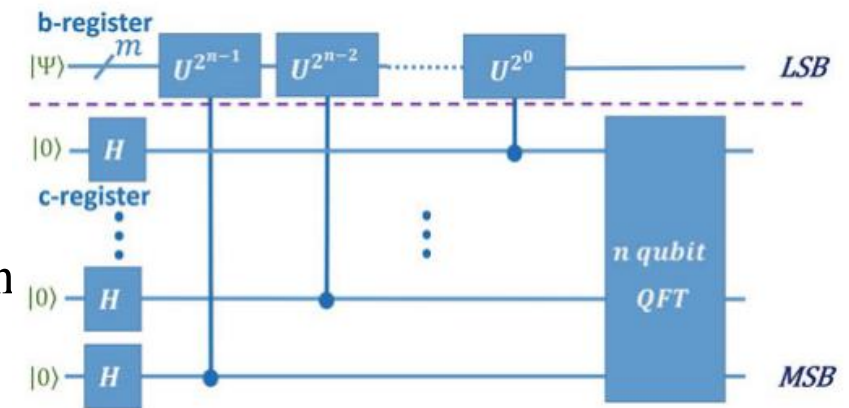
- Pic. 47 shows the implementation on IBM-Q. $|e_1\rangle$ is created by using a NOT gate.
- We can see the controlled-Z-gate that is a vertical line with two dots. There is no controlled- Z^2 gate because $Z^2 = I$. The I gate in the circuit is there just to align the circuits.
- The QFT circuit is only 2 qubits.
- You can see that the output is 100% $|100\rangle$. The first two qubits are for the c-register. Therefore, $|k\rangle = |10\rangle = |2\rangle$. That means $2\pi\psi = \frac{2\pi k}{2^n} = 2\pi 2/2^2 = \pi$, which is the phase of the corresponding eigenvalue, -1 .
- Note that we did not measure the b-register (the LSB); therefore, it is 0 by default in the classical register.



Pic. 47 Implementation of the QPE circuit. Note that the bottom is the MSB

General QPE Circuit

- Pic. 48 shows the general QPE circuit.
- We note that the b-register can be m -qubit that is required if we are finding the eigenvalue phase in a $m \times m$ matrix case. In that case, the controlled gate is a $m + 1$ -qubit gate (including the control bit).
- The c-register can be n qubits.
- n determines the accuracy of the phase it can estimate, larger n gives a higher resolution.
- The resolution is $1/2^n$, when $k = 1$. Usually, the ψ is not a fraction nor an integer as in the case we have discussed. Therefore, we cannot get the exact value, and thus it is an “estimation.” In summary, the QPE has $m + n$ qubits.
- The QPE seems not to be useful as we need to know the eigenvector (as input to the b-register) to find its phase. However, every vector is a linear superposition of the eigenvectors. Due to the linearity of quantum mechanics, their phases can be estimated at the same time. If we use it correctly, we can use it to solve other more important problems such as in the HHL algorithm for solving systems of linear equations.



Pic. 48 General implementation of the QPE circuit